Atlas Coach Intelligence Layer

The full PR roadmap — PR 1 through the frontier

Prime directive: the LLM is the voice, the code is the brain.

How to read this

One concern per PR. Each PR is built by Claude Code and goes through the Codex gate. The model tag on each card says which model to build it on; every Claude Code prompt opens with a model-switch hold point because the agent cannot change its own model.

How behavior is introduced (the safe ladder)

Preserve then read-only math then prescription (as a preview) then decisions then thin LLM explanation then autoregulation then personalization, memory, confidence then advanced. The engine does roughly 95% of the work; the LLM is added late and stays thin. The LLM never makes a progression or safety decision and never computes a number — that one line never moves.

Sequencing note — do not overtake the P0 lane

This roadmap slots **around** the P0 workout-reliability lane, not ahead of it. Finish P0 reliability first; pull in Coach Intelligence PRs as capacity allows. The early data PRs (PR 1, PR 2) are cheap and low-risk and can count toward the proving period's clean-PR run through the gate.

Rules that hold across every PR

- **One concern per PR**; if a diff is too big for the gate, split at the noted points.
- **Trust loop preserved**: preview then approve then write. No real Google Sheets writes without approval; dry-run first.
- **Never** expose secrets; **never** merge without approval.
- **Escalate to Dale** on schema, destructive, or governance decisions — the agent does not resolve those itself.
- **Model-switch hold point** in every Claude Code prompt; the agent stops, states the model, and waits.
- **The LLM/safety line never moves**: the LLM explains, the code decides; safety is the most conservative module and can veto.

Sonnet 4.6 = data / mechanical / docs	**Opus 4.8** = behavior-changing / correctness

Most Coach Intelligence PRs are Opus, because they introduce decisions or correctness-critical math. The data and ontology PRs are Sonnet.

At a glance

PR	Title	Model	Phase
1	Preserve + structure research (scaffold)	Sonnet	P0 Knowledge
2	Expand exercise ontology (4 to ~50)	Sonnet	P1 Foundations
3	IntensityModule: e1RM + RPE/%1RM (read-only)	Opus	P1 Foundations
4	VolumeModule: weekly sets per muscle (read-only)	Opus	P1 Foundations
5	SafetyModule (classifier) + onboarding health screen	Opus	P1 Foundations
6	Per-user State / Feature object (read-only substrate)	Opus	P1 Foundations
7	Starter program templates (state machines)	Opus	P2 Core (MVP)
8	ProgressionModule (double progression + beginner LP)	Opus	P2 Core (MVP)
9	Cold-start onboarding flow	Opus	P2 Core (MVP)
10	Thin LLM explanation layer (v1)	Opus	P2 Core (MVP)
11	Readiness check-ins + ReadinessModule	Opus	P3 Recovery
12	FatigueModule (recovery clocks + acute:chronic load)	Opus	P3 Recovery
13	DeloadModule	Opus	P3 Recovery
14	ExpectedPerformance + PlateauModule	Opus	P3 Recovery
15	Autoregulation upgrade to ProgressionModule	Opus	P3 Recovery
16	SubstitutionModule	Opus	P4 Personalize
17	Goal / population templates & caps (+ 5/3/1)	Opus	P4 Personalize
18	Memory architecture (trends + coach memory + entity resolution)	Opus	P4 Personalize
19	ConfidenceModule (ask vs act)	Opus	P4 Personalize
20	NutritionModule (minimum viable + guardrails)	Opus	P5 Nutrition/Comms
21	Behavior-change & communication layer	Opus	P5 Nutrition/Comms
22	Periodization & peaking engine	Opus	P6 Frontier
23	VBT support (optional)	Opus	P6 Frontier
24	Wearables + outcome learning (frontier)	Opus	P6 Frontier

Amber underline marks a milestone: **PR 10** working beginner coach (MVP), **PR 15** adaptive coach, **PR 19** personalized coach, **PR 21** complete core coach.

PHASE 0 · Knowledge Foundation

Preserve the research as structured, provenance-tracked knowledge. No behavior.

PR 1 — Preserve + structure research (scaffold)	Sonnet 4.6 data / mechanical
Concern	Ingest the 4 research docs as an immutable archive + human digest + structured JSON config (schemas, exercise ontology with 4 worked examples, evidence tiers, progression/fatigue/safety rules). Provenance on every entry.
Depends on	None.
Guardrails	Static data + docs only. No module consumes any config. No behavior, no LLM wiring, no Sheets / user-data access.
Done when	Trees exist under docs/research/coaching-intelligence/ and config/coaching/; 4 exercises validate against the schema; PDFs placed in the archive; Codex green.

PHASE 1 · Read-only Foundations

Build modules that READ existing logs + the new config and COMPUTE values. No recommendations, no write-path changes.

PR 2 — Expand exercise ontology (4 to ~50)	Sonnet 4.6 data / mechanical
Concern	Populate the ontology to ~50-60 common lifts following the existing schema and _index; keep movement patterns and substitution links consistent.
Depends on	PR 1.
Guardrails	Data only. No consumer. No behavior. Dangling substitution ids still acceptable but should shrink.
Done when	~50 validated exercise files; _index updated; Codex green.

PR 3 — IntensityModule: e1RM + RPE/%1RM (read-only)	Opus 4.8 behavior / correctness
Concern	Compute e1RM per lift from existing logs (Epley/Brzycki + averaging) and an RPE-to-%1RM lookup. Pure read; surfaces values only.
Depends on	PR 1; existing Log/Effort data.
Guardrails	Read-only over logs. Makes NO recommendation. No write path touched. e1RM treated as an estimate with error, per the caveats.
Done when	e1RM + lookup unit-tested against reference values; no writes; Codex green.

PR 4 — VolumeModule: weekly sets per muscle (read-only)	Opus 4.8 behavior / correctness
Concern	Tally hard sets per muscle and per pattern per week from logs + ontology; compare against the landmark config. Read-only.
Depends on	PR 1, PR 2, PR 3.
Guardrails	Read-only. Landmarks are heuristic ranges, surfaced as such. No recommendation, no writes.
Done when	Volume tallies unit-tested against fixtures; Codex green.

PR 5 — SafetyModule (classifier) + onboarding health screen	Opus 4.8 behavior / correctness
Concern	Read safety.rules.json + the injury profile + symptom input and output a green/yellow/red flag; add a PAR-Q-style onboarding screen. The guardian layer.
Depends on	PR 1.
Guardrails	Conservative defaults; never diagnose; red flag means stop + refer; low confidence means MORE caution. Built to later veto other modules. Never coach through.
Done when	Red-flag classification + screen tested; stop/refer paths verified; Codex green.

PR 6 — Per-user State / Feature object (read-only substrate)	Opus 4.8 behavior / correctness
Concern	Assemble a compact per-user, per-lift state: e1RM trend, volume vs landmarks, PRs, staleness, adherence. The substrate that decisions and the LLM will read.
Depends on	PR 3, PR 4.
Guardrails	Derived / read-only. Stores rolling summaries, not a duplicate of the raw log. No recommendation yet.
Done when	State object unit-tested; recomputation deterministic; Codex green.

PHASE 2 · Core Coaching Loop (MVP)

The first real coach: onboard, prescribe, progress, explain. Behavior is introduced here, as a preview the user accepts.

MILESTONE at PR 10: a working beginner coach.

PR 7 — Starter program templates (state machines)	Opus 4.8 behavior / correctness
Concern	Encode StrongLifts 5x5 and GZCLP as data plus a deterministic template runner that produces the next prescribed session.
Depends on	PR 1, PR 3.
Guardrails	Prescription is a PREVIEW the user accepts (trust loop: preview, approve, write). No silent Sheets writes. Templates are data + a runner only.
Done when	Runner produces the correct next session for fixtures; trust loop preserved; Codex green.

PR 8 — ProgressionModule (double progression + beginner LP)

Opus 4.8
behavior / correctness

Concern	Consume progression.rules.json + the state object to decide increase / hold / reduce / add-reps for the next session. The decision core.
Depends on	PR 3, PR 6, PR 7.
Guardrails	Decisions deterministic + testable; never invent increments outside config; defers to SafetyModule. Preview, not a silent write.
Done when	Decision table covered by tests including fail and deload paths; Codex green.

PR 9 — Cold-start onboarding flow

Opus 4.8
behavior / correctness

Concern	Questionnaire to population-prior defaults to an assigned starter template + conservative training maxes (the 90% rule). Decision-tree, no LLM.
Depends on	PR 5, PR 7, PR 8.
Guardrails	Conservative defaults; no LLM math; the safety screen runs first.
Done when	A new-user path yields a safe starting program; Codex green.

PR 10 — Thin LLM explanation layer (v1)

Opus 4.8
behavior / correctness

Concern	A function-calling interface exposing the engine (getNextWorkout, getProgressionDecision, getVolumeStatus, checkSafety). The LLM only explains engine output and adds light encouragement. Cached system prompt; structured I/O.
Depends on	PR 5, PR 8.
Guardrails	The LLM NEVER computes numbers, NEVER overrides the engine, ALWAYS defers to SafetyModule. No secrets in the prompt. This is the line that never moves.
Done when	LLM explanations match engine outputs; an adversarial test shows it will not invent loads; Codex green.

PHASE 3 · Recovery & Autoregulation

Turn the tracker into an adaptive coach: read readiness, manage fatigue, deload, spot plateaus, autoregulate.

MILESTONE at PR 15: an adaptive coach.

PR 11 — Readiness check-ins + ReadinessModule

Opus 4.8
behavior / correctness

Concern	Capture sleep / soreness / stress / motivation; compute a subjective-weighted readiness score.
Depends on	PR 6.
Guardrails	Subjective-weighted; any wearable input is supporting only, with wide error bars. Readiness is a soft signal, not a verdict.
Done when	Score computed + tested; Codex green.

PR 12 — FatigueModule (recovery clocks + acute:chronic load)	Opus 4.8 behavior / correctness
Concern	Per-muscle recovery clocks + an EWMA acute:chronic load ratio from logs.
Depends on	PR 4, PR 6.
Guardrails	Recovery priors are heuristic / low-confidence; ACWR is a monitoring flag, not a guarantee.
Done when	EWMA + clocks unit-tested; Codex green.

PR 13 — DeloadModule	Opus 4.8 behavior / correctness
Concern	Planned and auto-triggered deloads from fatigue / readiness / performance (fires when 2 or more triggers agree).
Depends on	PR 11, PR 12.
Guardrails	Conservative, explainable triggers; a deload is a preview with user-visible reasoning.
Done when	Trigger logic tested for both planned and auto paths; Codex green.

PR 14 — ExpectedPerformance + PlateauModule	Opus 4.8 behavior / correctness
Concern	Predict today's likely output; detect plateau vs noise; run the underperformance differential (sleep / fatigue / injury / illness / motivation / programming).
Depends on	PR 6, PR 11, PR 12.
Guardrails	Ignore a single bad session unless red flags; flag a plateau only across the defined window with adequate recovery. Confirmed plateaus go to coach memory.
Done when	Plateau vs noise tested on fixtures; the differential mapping verified; Codex green.

PR 15 — Autoregulation upgrade to ProgressionModule	Opus 4.8 behavior / correctness
Concern	Set loads from current e1RM / readiness rather than a fixed template; the graduate-from-LP logic.
Depends on	PR 8, PR 11, PR 14.
Guardrails	Still a preview; still defers to safety; per-user RPE calibration where the data allows.
Done when	The autoregulation path is tested against LP; Codex green.

PHASE 4 · Personalization, Memory & Confidence

Make it feel like a real coach: reference history, adapt to goal and context, and ask smart questions instead of guessing.

MILESTONE at PR 19: a personalized coach.

PR 16 — SubstitutionModule

Opus 4.8
behavior / correctness

Concern	Swap exercises by pattern / muscle / equipment / fatigue similarity; equipment-aware filtering; add a referential-integrity check on the now-fuller ontology.
Depends on	PR 2, PR 4.
Guardrails	Substitutions preserve the stimulus; respect contraindications and the safety layer.
Done when	Substitution matching tested; integrity check added; Codex green.

PR 17 — Goal / population templates & caps (+ 5/3/1)

Opus 4.8
behavior / correctness

Concern	Powerlifting / bodybuilding / general / weight-loss templates; older-adult / youth / busy-parent / home-gym caps; add 5/3/1 for intermediates.
Depends on	PR 7, PR 8, PR 15.
Guardrails	Population caps conservative; templates are data + a runner; goal is user-chosen or detected from the training lean.
Done when	Each profile yields an appropriate program; caps enforced; Codex green.

PR 18 — Memory architecture (trends + coach memory + entity resolution)

Opus 4.8
behavior / correctness

Concern	A derived long-term-trends store + coach memory (effective tone, sensitivities like sleep-sensitive bench) + entity resolution (low-bar = back squat).
Depends on	PR 6, PR 14.
Guardrails	Store facts and decisions; infer judgments. A per-user derived layer, not edits to the universal KB. No sensitive overreach.
Done when	Trends recomputed deterministically; entity resolution tested; Codex green.

PR 19 — ConfidenceModule (ask vs act)

Opus 4.8
behavior / correctness

Concern	Score confidence (completeness / recency / consistency / self-report reliability); thresholds for act, act-with-caveat, or ask; wire the LLM to ask a clarifying question when confidence is low.
Depends on	Most prior modules.
Guardrails	Safety inverts the logic: low confidence about a red flag triggers MORE caution. Ask only when genuinely lacking info, so questions feel earned.
Done when	Confidence gating tested across high / medium / low; the LLM asks appropriately; Codex green.

PHASE 5 · Nutrition & Communication

Round out the coach with minimum-viable nutrition and durable, non-repetitive coaching language.

MILESTONE at PR 21: a complete core coach.

PR 20 — NutritionModule (minimum viable + guardrails)

Opus 4.8
behavior / correctness

Concern	Protein and calorie targets, rate-of-change guardrails, and an evidence-ranked supplement reference.
Depends on	PR 1 (evidence tiers); profile bodyweight and goal.
Guardrails	HARD limits: no clinical diets, no treating medical conditions, no disordered-eating counsel; refer to a dietitian / physician; flag eating-disorder warning signs.
Done when	Targets computed; refer-out paths verified; Codex green.

PR 21 — Behavior-change & communication layer

Opus 4.8
behavior / correctness

Concern	Adherence tracking, anti-repetition constraints, identity / encouragement language, and nudge timing.
Depends on	PR 10, PR 18.
Guardrails	Autonomy-supportive; specific and history-referencing, not generic; missing one session is not failure (anti-guilt). Do not foster over-reliance.
Done when	Anti-repetition + adherence nudges tested; Codex green.

PHASE 6 · Advanced / Frontier

Only after the fundamentals are trustworthy. Each is optional and gated on clear value over the simpler core.

Frontier: PR 22-24, after the core is solid.

PR 22 — Periodization & peaking engine

Opus 4.8
behavior / correctness

Concern	Block / DUP scheduling; a separate strength-sport peaking and taper engine kept out of the general-population logic.
Depends on	PR 15, PR 17.
Guardrails	Peaking only for the right population; general users get consistency, not peaking.
Done when	Schedules generated for test athletes; Codex green.

PR 23 — VBT support (optional)

Opus 4.8
behavior / correctness

Concern	A velocity field + an individual load-velocity profile regression for autoregulation and velocity-loss cutoffs.
Depends on	PR 15.
Guardrails	Load-velocity profiles vary by device and exercise; individual MVT beats population values; wide limits of agreement acknowledged. Optional.
Done when	LVP regression + a daily-load picker tested; Codex green.

PR 24 — Wearables + outcome learning (frontier)	Opus 4.8 behavior / correctness
Concern	Optional HRV / sleep ingestion as supporting signals; learn individual dose-response and recovery from accumulated data.
Depends on	PR 11, PR 12, PR 18.
Guardrails	Wearables supporting only; respect upstream API AI-use terms (Oura, Strava); privacy and consent; learned params are noisy, so use them qualitatively. Last, not first.
Done when	Ingestion + learning behind flags; clear value vs the simpler logic; Codex green.

Where this lands

When the core is done (through PR 21), Atlas is a real coaching system, not a logger with a chatbot. A new user is screened for safety, onboarded with a sensible starter program, and coached session to session with deterministic progression. As history accumulates, Atlas reads readiness, manages fatigue, deloads on real signals, spots plateaus, autoregulates load, and personalizes by goal and life context. It references the user's own history, asks a clarifying question only when it genuinely lacks information, and explains every decision in plain language. The LLM is the voice throughout; the code is the brain.

> **What would change this plan**
>
> If P0 reliability needs attention, pause the Coach Intelligence lane — reliability wins. If Codex flags a PR as too large, split it at the noted points. If a module's heuristics (volume landmarks, recovery clocks, readiness weights) produce complaints, lower the defaults — they are individual, and the user's own data eventually beats the population priors. And if anyone ever proposes letting the LLM make a progression or safety decision to save effort: that is the one architectural line that does not move.